

Kina Kim  
July 25th, 2018  
Computer Museum

# Augmented Reality Error Documentation for Unity & Vuforia

# General Setting & Tips to get around problems:

1. Download Unity
  - a. Make sure to implement “Vuforia Augmented Reality” by putting a check box when given the options.
2. Edit -> Project Settings -> Player -> Inspector (not Services) -> XR Settings  
-> tick on “vuforia Augmented Reality Support”
  - a. There are options of PC, Mac & Linux/iphone/android/UWP. If already decided, choose one of the platform and make such changes on that tab. Else, keep it on PC, Mac & Linux set up.
  - b. Then Vuforia folder appears on Project
  - c. Specific to Android, go to “Other Settings”. Deselect Android TV Compatibility.
    - i. “Vuforia does not support Android TV” warning will disappear
3. Right click on Hierarchy,  
Vuforia -> AR Camera
  - a. It will ask you to import. Click import.
  - b. Another way of implementing AR files is by downloading Vuforia-Unity-6-2-10 at Vuforia Developer website. **NEVER DO THIS.**
    - i. The files inside have error and right after you implement it into your unity, it will start crashing saying that there’s **compilation error**.
4. Delete “Main Camera” by clicking right click.
  - a. Orelse, there’s two cameras...
5. AR Camera -> Open Vuforia configuration -> App License Key
  - a. On there, you add the app license key => will elaborate further when explaining Vuforia
6. Cloud Recognition -> Access Key & Secret Key
  - a. Will be bringing from Vuforia as well.

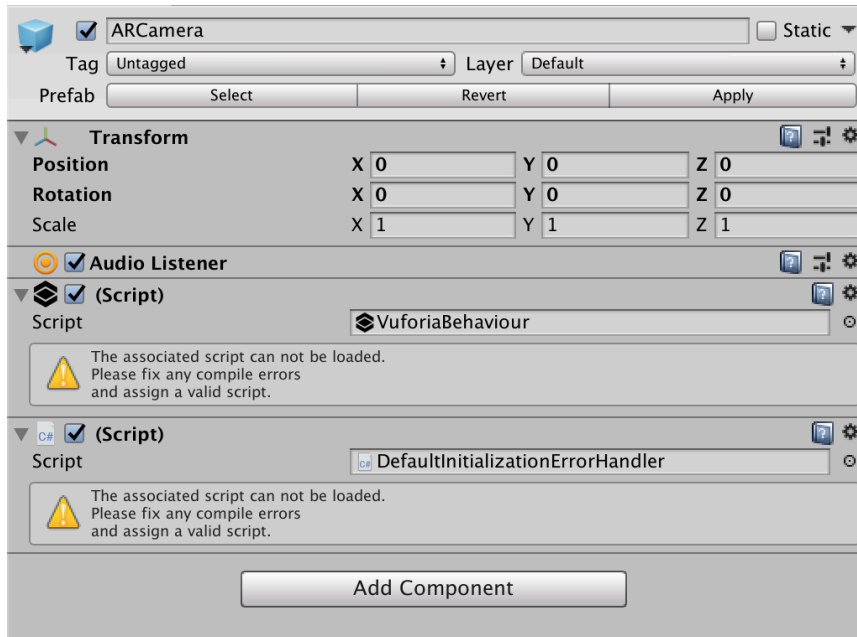
Please watch “[General Setting](#)” video for visual representation and more detailed steps.

=> If the hyperlink doesn’t work, please look for “General Setting” video in the same folder as where you found this document.

# Errors I have faced:

## Errors while building

1. Compile errors from all the scripts after implementing unity vuforia package

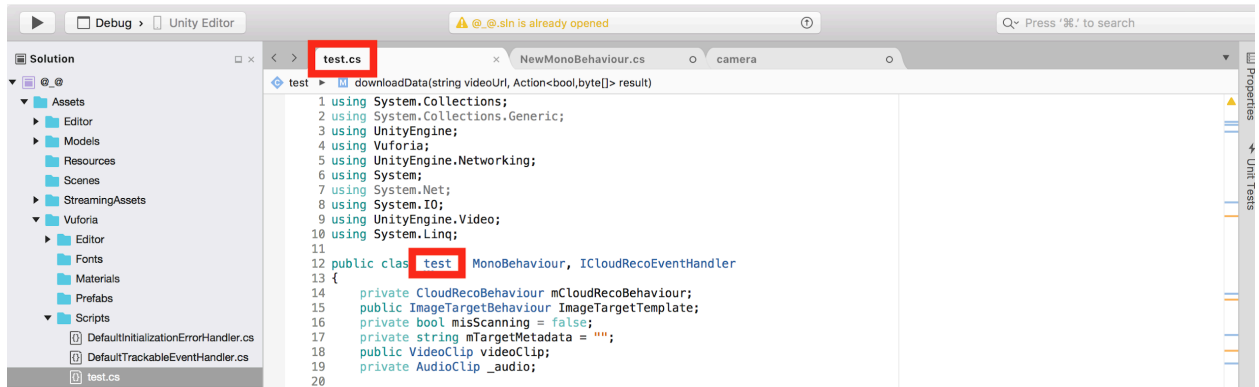


For every single scripts, this may give an error “The associated script can not be loaded”  
This happened to me when I dragged in the “Vuforia-unity-6-2-10”

“Possible” ways to fix:

- No certainty because this method had so many issues that I used another way
  - a. Re-download Unity
  - b. Re import the unity vuforia package

## 2. Class name matching with the script name

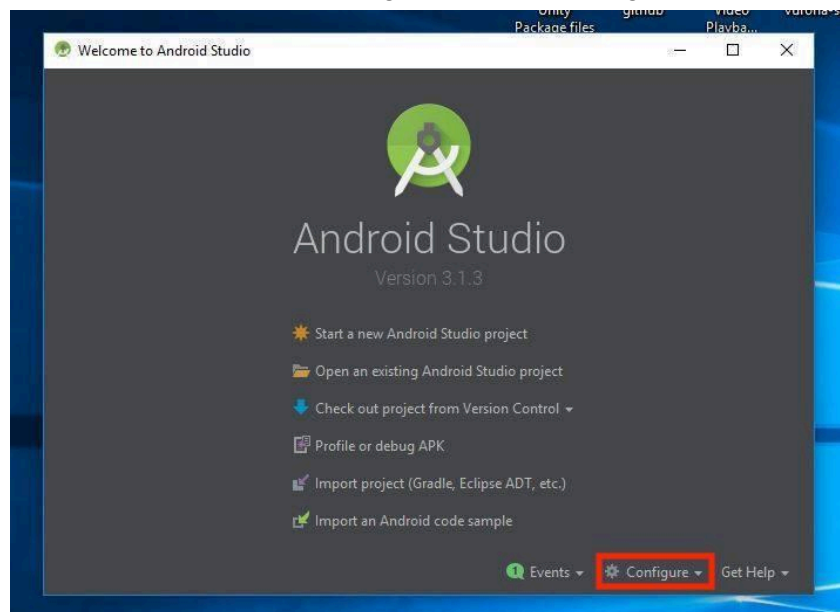


- a. You will get a compile error.
    - i. This is different from the previous error. It won't give compile error from EVERY SINGLE SCRIPT. Plus, it's more likely for this to happen if it's the script that you have built.
  - b. This may block you from dragging the script into any function you need this script.
- ## 3. A lotttttttttttttttttt of common errors found while coding
- a. Syntax error
  - b. Obsoletes
  - c. AND MORE...
    - i. "Just believe that it will work one day and continue working through"

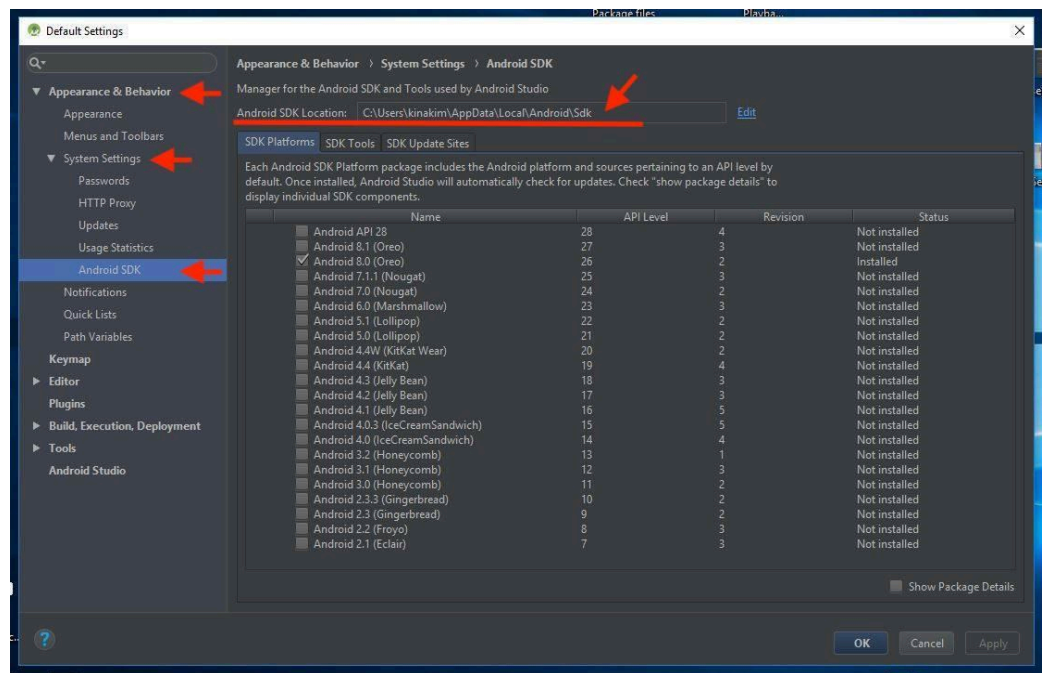


## How to & Errors while uploading Unity onto your device

4. Switch platform specific to the device.
  - a. I've used Android. After clicking on "Android", click switch platform on the bottom.
5. Unity finding the path way of SDK file
  - a. Edit -> preferences -> External -> need to set up the pathway to SDK and JDK
    - i. **DON'T get the latest** JDK. Download **jdk 1.8.0\_181**.
    - ii. Download Android Studio for SDK.
6. Having hard time finding the pathway of JDK or SDK folder.
  - a. For JDK
    - i. Normally gets saved at:
      1. This PC -> Local Disk (C:) -> Program Files -> Java
        - a. You need to direct the path to one specific folder in this Java folder
  - b. For SDK
    - i. Download Android Studio
    - ii. Open Android Studio -> Configure -> SDK Manager

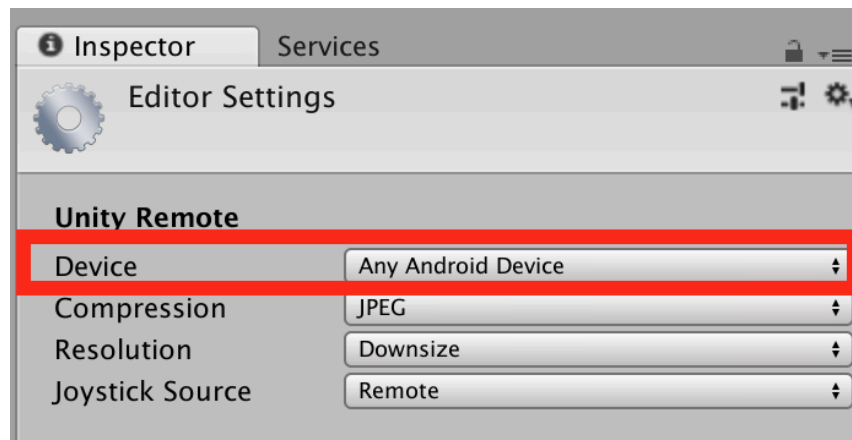


iii. SDK Manager -> copy the Android SDK location

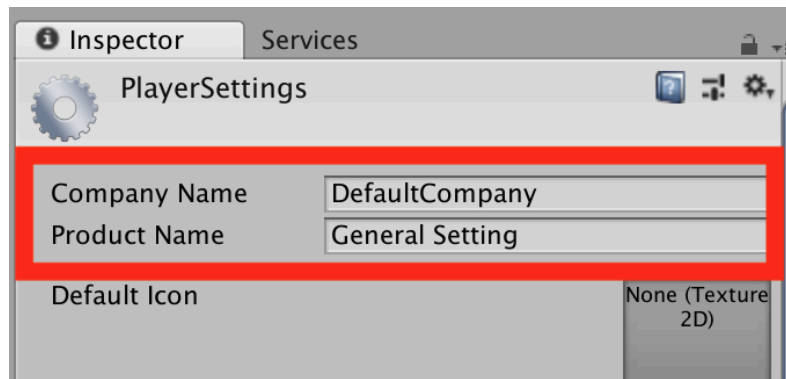


7. Edit -> Project Settings -> Editor -> Device -> Any Android device

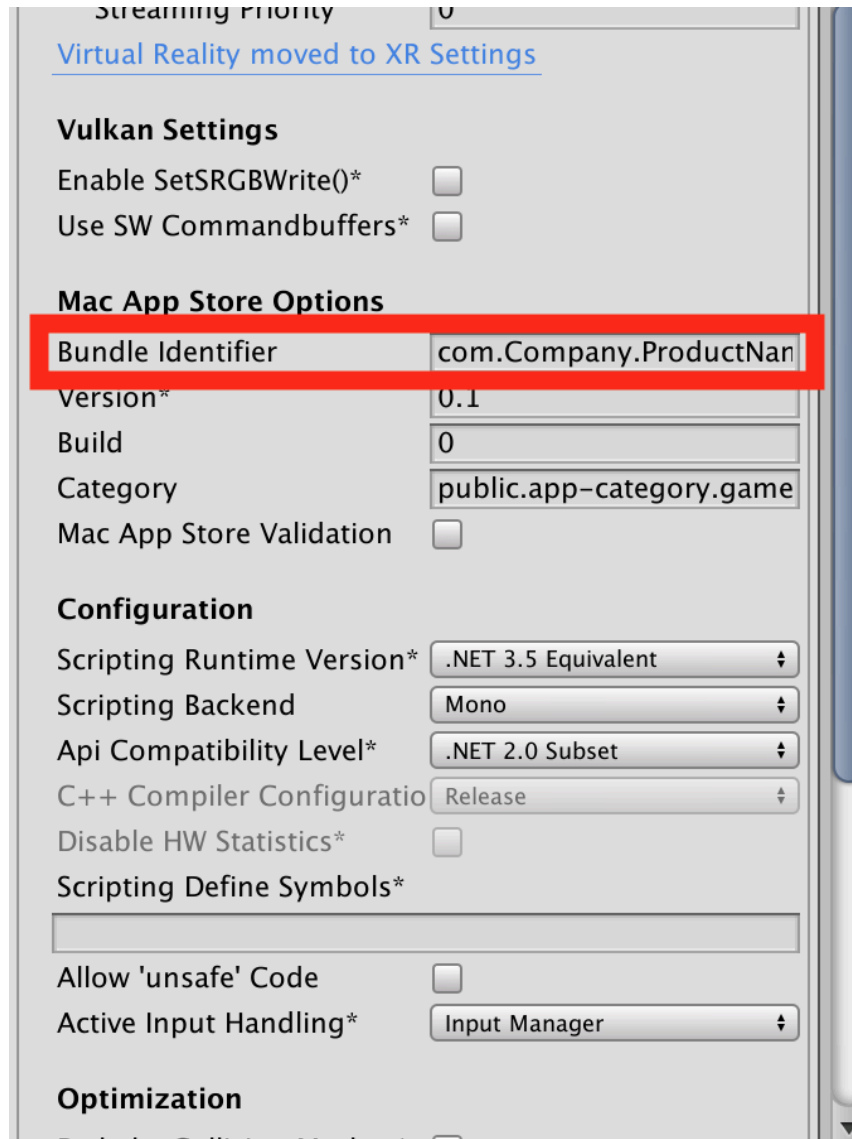
a.



8. Build Setting -> Player Settings
  - a. Change Company Name and Product Name



- b. Scroll down. On "Other settings" Make sure to also change them on "Mac App Store Options".



## 9. Back to “Build Settings”

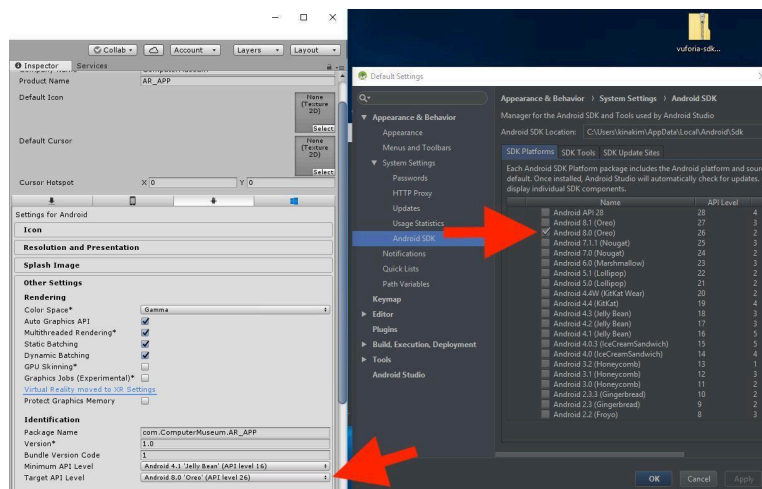
- Connect your Android device to the computer.
- Click Build And Run.

## 10. “Unable to list target platforms. Please make sure the android sdk path is correct” error

- Please follow this youtube guide.  
<https://www.youtube.com/watch?v=xH3lyvtsLuc>

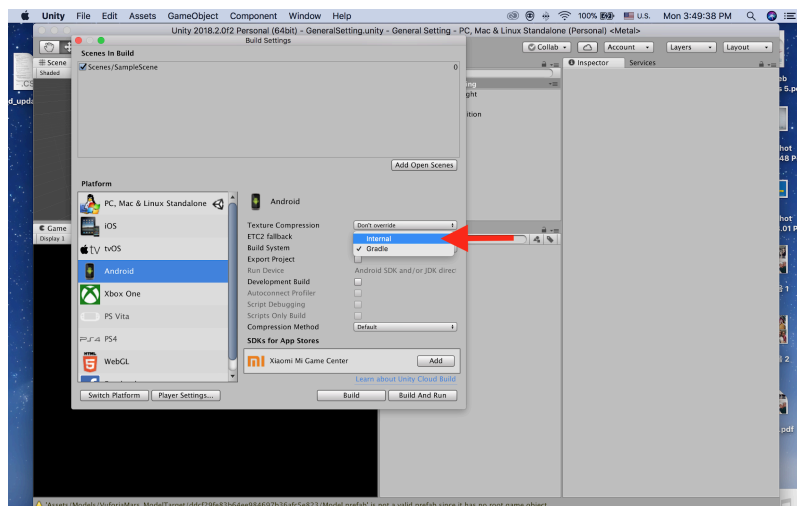
## 11. "SDK Plamform Tools version 0.0.0<23.0.0" error

- Make sure that Target API Level on “Player Settings” match with the Android SDK you have in your Android



## 12. “Gradle build failed” error

- Change Build system to internal.



## 13. Hit “Build and Run” again. It should work!!



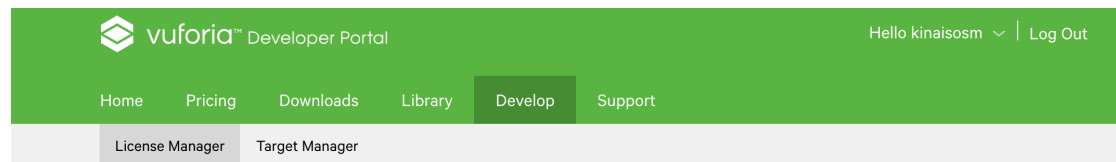
# How to handle Vuforia:

## 1. Go to vuforia website

- Type in “Vuforia development” in google. The first one is the site that you have to enter.
- Just “Vuforia” is directs you to a different website. This cannot be used for a development purpose.
- User name: [computer.museum@uwaterloo.ca](mailto:computer.museum@uwaterloo.ca)

Password: UWComputerMuseum55

## 2.



### License Manager

Create a license key for your application.

Get Development Key

Buy Deployment Key

| Name                              | Type    | Status ▾ | Date Modified      |
|-----------------------------------|---------|----------|--------------------|
| <a href="#">AR first trial</a>    | Develop | Active   | Jul 09, 2018 15:28 |
| <a href="#">CLOUD recognition</a> | Develop | Active   | Jul 13, 2018 14:20 |
| <a href="#">CouldRecognition</a>  | Develop | Active   | Jul 11, 2018 09:16 |

- Go to “Develop” tab and License Manager
- Click “Get Development Key” to create new license.
  - Can only get up to 1000 images uploaded on each license
- Then such screen will pop up. Then you can create a new license key.

### Add a free Development License Key

#### App Name

You can change this later

#### License Key

Develop

Price: No Charge

Reco Usage: 1,000 per month

Cloud Targets: 1,000

VuMark Templates: 1 active

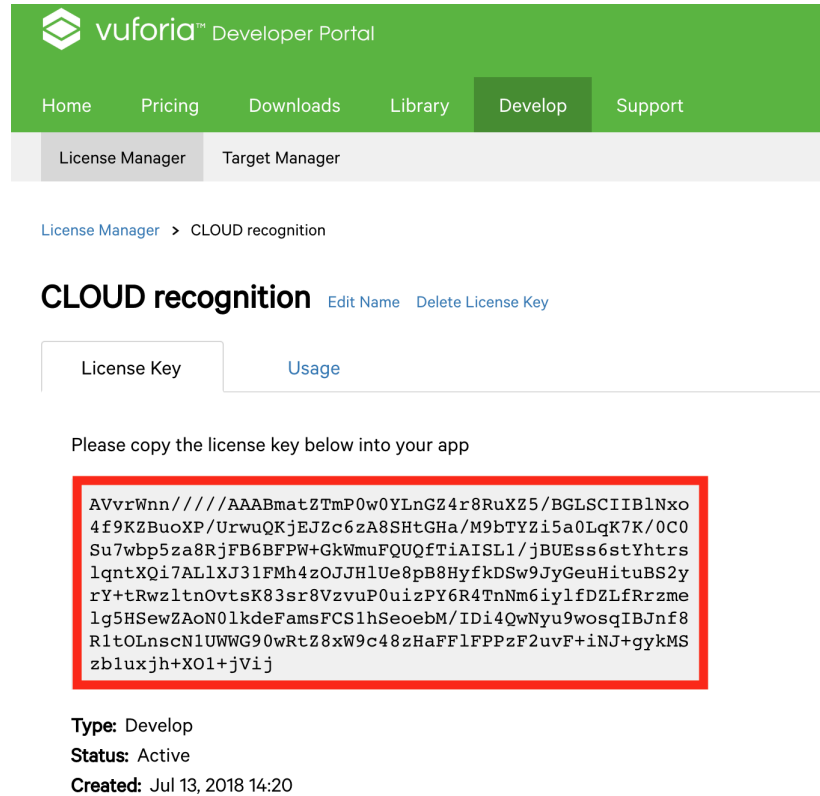
VuMarks: 100

☐ By checking this box, I acknowledge that this license key is subject to the terms and conditions of the [Vuforia Developer Agreement](#).

Cancel

Confirm

- d. Once you create one, you click into the page that looks like the one below by clicking into the name of the license that you've created. Copy the texts in the red box that I've indicated.



**vuforia™ Developer Portal**

Home Pricing Downloads Library **Develop** Support

License Manager Target Manager

[License Manager](#) > CLOUD recognition

### CLOUD recognition

[Edit Name](#) [Delete License Key](#)

License Key [Usage](#)

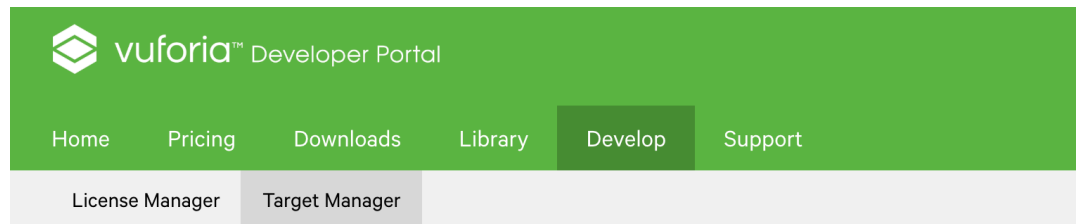
Please copy the license key below into your app

```
AVvrWnn/////AAABmatZTmP0w0YLnGZ4r8RuXZ5/BGLSCIIB1Nxo
4f9KZBuoXP/UrwuQKjEJZc6zA8SHtGHa/M9bTYZi5a0LqK7K/0C0
Su7wbp5za8RjFB6BFPW+GkWmuFQUQfTiAISL1/jBUEss6stYhtrs
lqntXQi7ALlXJ31FMh4zOJJH1Ue8pB8HyfkDSw9JyGeuHituBS2y
rY+tRwz1tnOvtsK83sr8VzvuP0uizPY6R4TnNm6iy1fDZLfRrzm
lg5HSewZAoN0lkdeFamsFCS1hSeoebM/IDi4QwNyu9wosqIBJnf8
R1tOLnscN1UWWG90wRtZ8xW9c48zHaFFlFPPzF2uvF+iNJ+gykMS
zbluxjh+X01+jVij
```

**Type:** Develop  
**Status:** Active  
**Created:** Jul 13, 2018 14:20

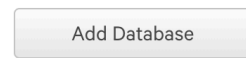
- e. Then you can add this to Unity.

3. Now we are going to add targets into that license key created.
  - a. Click into Target Manager tab. Then click into “Add Database”.



## Target Manager

Use the Target Manager to create and manage databases and targets.



- b. Then this screen will pop out. Make sure to click “Cloud” which will allow you to use cloud recognition. Then create a database by filling in the necessary informations.

The image shows a modal window titled "Create Database". It has a white background and is overlaid on a dark gray background. The form contains the following elements: a "Name:" label followed by a text input field; a "Type:" label followed by three radio button options: "Device", "Cloud", and "VuMark". The "Cloud" option is selected and highlighted with a red rectangular box. Below the radio buttons is a dropdown menu labeled "Select License Key" with a downward arrow. Under the dropdown, there is a line of text: "To create a license key please go to the [License Manager](#)". At the bottom right of the modal are two buttons: "Cancel" and "Create".

- c. For example, I have named my database as “CLOUD”. Click into that name to add targets into that specific database.

 vuforia™ Developer Portal

[Home](#)[Pricing](#)[Downloads](#)[Library](#)[Develop](#)[Support](#)

[License Manager](#)[Target Manager](#)

## Target Manager

Use the Target Manager to create and manage databases and targets.

Add Database

| Database              | Type  | Targets |
|-----------------------|-------|---------|
| <a href="#">CLOUD</a> | Cloud | 4       |

- d. Once clicked, this screen will appear. Then click “Add Target”.

**CLOUD** [Edit Name](#)

Type: Cloud

License Key: [CLOUD recognition](#)

Targets (4)

[Database Access Keys](#)

Add Target

Search by target name or target ID

- e. This screen will pop up. This will allow you to add the target.

The screenshot shows a modal dialog box titled "Add Target". It contains the following fields and controls:

- Target Image File:** A section with a "Choose File" button and a "Browse..." button. Below these is the text ".jpg or .png (max file 2mb)".
- Width:** A text input field. Below it is a paragraph of instructions: "Enter the width of your target in scene units. The size of the target should be on the same scale as your augmented virtual content. Vuforia uses meters as the default unit scale. The target's height will be calculated when you upload your image."
- Metadata Package:** Labeled as "(Optional)", it includes a "Browse..." button and the text "max file 2mb".
- Name:** A text input field.
- At the bottom right are "Cancel" and "Add" buttons.

- i. Make sure to follow the instruction as thoroughly as possible.
- ii. JPG or PNG file only for target image
- iii. Width - I've done 5 and seemed to work fine.
- iv. Metadata package is an associated information that pop up when the particular image is scanned. You can attach this file by clicking "Browse" on Metadata package. You will have to have metadata that you want to be saved somewhere in your computer to attach it. Make sure the file is less than 2mb. It works the best if you have the metadata saved on Notepad.
  1. IF want to have video/audio played, save the url ending either with mp4 or mp3 as metadata text.
- v. Once completed the steps, click "Add".

- vi. Then you will see that the target has been created. It will have “processing” as a status.

## CLOUD [Edit Name](#)

Type: Cloud

License Key: [CLOUD recognition](#)

Targets (4)

[Database Access Keys](#)

Add Target

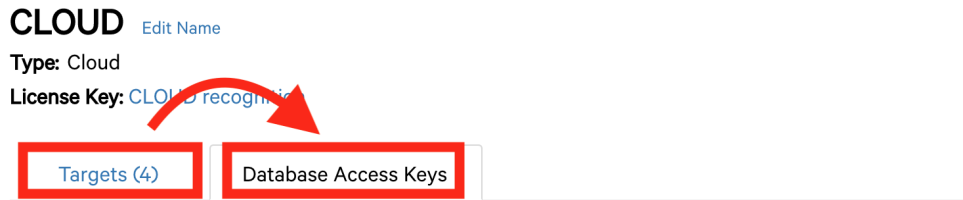
Search by target name or target ID

| <input type="checkbox"/> | Target Name                                                                                       | Rating | Recos ▾ | Status ▾ | Date Modified ▾    |
|--------------------------|---------------------------------------------------------------------------------------------------|--------|---------|----------|--------------------|
| <input type="checkbox"/> |  Capture_offical | ★★★★★  | 38      | Active   | Jul 20, 2018 10:11 |
| <input type="checkbox"/> |  UFO             | ★★★★★  | 21      | Active   | Jul 19, 2018 14:23 |
| <input type="checkbox"/> |  SC              | ★★★★★  | 36      | Active   | Jul 19, 2018 14:20 |
| <input type="checkbox"/> |  JustTrying      | ★★★☆☆  | 89      | Active   | Jul 17, 2018 15:24 |

Last updated: Today 10:28 AM [Refresh](#)

1. Underneath, click on “Refresh” until it becomes “Active” status like shown above.
2. The stars next to the image shows how well it can be scanned. It’s a rating.
3. You can click into that image by clicking its name. There, you can delete the metadata and upload a new one to update them.

#### 4. Connecting these target images to Unity



Access Keys are specific to each cloud database. You should never share your access keys with an untrusted party and should take the necessary steps to protect them within your application code.

##### Client Access Keys

Access Key: 80ecfdc60ee01172fb9c20f8c47ff862f6fbd060

Secret Key: 7b90337b9a2bd92012bc6c5cd1999731f1325f0a

The Client Access Keys must be passed to the Vuforia library within the app in order to authenticate itself with the server

##### Server Access Keys

Access Key: 25d77498b2d9c47551e85874ffff37090c26c877

Secret Key: 665c619e3e4577dea7789d9f8ec808422c6de94b

Use the Server Access Keys to upload and manage images in your cloud target database, via the REST interfaces

- Go to “Database Access Keys” which is right new to the “Targets” tab that you are in.
- You can get Access Key and Secret Key here. You must copy those keys from “**Client Access**”.
- These will be copied down to “Cloud Recognition” hierarchy in Unity.

Please watch “[Unity & Vuforia](#)” video for visual representation and more detailed steps for connecting Unity and Vuforia.

=> If the hyperlink doesn't work, please look for “Unity & Vuforia” video in the same folder as where you found this document.

# How to do Collaboration with others:

Please watch “[Collab](#)” video for visual representation and more detailed steps.

=> If the hyperlink doesn't work, please look for “Collab” video in the same folder as where you found this document.

## Next steps forward:

1. Rescale the information boxes that it's readable on devices.
  - a. It will be great if the information boxes can show up based on the proportion calculated in the code that on every Android device, the users won't have difficult time reading.
2. Change the colors of the information boxes
  - a. Maybe based on three basic colors representing computer museum
    - i. Red
    - ii. Black (We can keep the same box as now)
    - iii. Yellow
3. Create a main menu screen that leads to the scanning screen
4. Create a button that you can press to exit the app
5. Figure out a way to upload this app into App Stores
  - a. Originally planned to create QR code which a person can scan and will lead to the app store. Yet, realized that in order to do so they will have to download a camera that can scan QR code. Needing two applications to download, people will be too bothered to actually try using Augmented Reality.
6. Create a poster explaining step by step how to use Augmented Reality from how to install to how to use it in this exhibition